JIANGMING LIU

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EMPLOYMENT

Associate Professor Yunnan University, China 2023 – Present
My research interests include natural language processing, deep learning algorithms, large language models, its associated computational linguistics and its downstream applications.

Senior Research Scientist Tencent, China 2021 – 2023
Research includes large language models, and recommendation and advertising systems.

Research Assistant Singapore University of Technology and Design, Singapore 2015 – 2017
Research includes syntactic parsing and text generation.

EDUCATION

DPhil computer science University of Edinburgh, UK Sep. 2017 – May 2021
Transmodal H2020 fund by European Research Council and Bloomberg fund (full scholarship) for DPhil research.
Thesis title: Understanding and generating language with discourse representation structures.

MSc computer science Beijing Jiaotong University, China Sep. 2012 – Mar. 2015
National Scholarship for Graduate Students.
Thesis title: Hierarchical Phrase-Based Statistical Machine Translation Incorporating Japanese Case Grammar

BEng computer science Beijing Jiaotong University, China Sep. 2008 – Jun. 2012
Scholarship from Beijing jiaotong University.

TEACHING

Operating Systems and Experiments	2023, 24, 25, 26. Fall	undergraduate
Software Development Process	2023, 24, 25. Fall	undergraduate
Prompt Engineering	2026. Fall	undergraduate

RESEARCH

Grants:

PI. National Natural Science Foundation of China. The Techniques of Discourse Representation Parsing for Multi-turn Dialogue Generation (¥330K). Jan. 2026 – Dec. 2029.
PI. Yunnan Province Science and Technology Department grant. Research on Large-Scale Language Models for Multi-Dimensional Securities Market Analysis (¥100K). Mar. 2024 – Feb. 2027.
PI. Yunnan Province Science and Technology Department grant. A large-scale language model based on heterogeneous knowledge (¥700K). Jan. 2024 – Dec. 2028.
PI. Yunnan University grant. Donglu Talent Program (¥1M). Feb. 2023 – Jan. 2029.

Research Students:

Hao Chen	MSc	advisor (Sep. 2026 – .)
Lihong Cheng	MSc	advisor (Sep. 2026 – .)
Wenhao Li	MSc	advisor (Sep. 2026 – .)
Pengfei Liang	MSc	advisor (Sep. 2026 – .)
Liying Wang	MSc	advisor (Sep. 2026 – .)
Longfei Fan	MSc	advisor (Sep. 2025 – .)
Shengjie Dong	MSc	advisor (Sep. 2025 – .)
Shifeng Xie	MSc	advisor (Sep. 2025 – .)
Xingming Yao	MSc	advisor (Sep. 2025 – .)
Guangbin Zhang	MSc	advisor (Sep. 2024 – .)
Yulin Dou	MSc	advisor (Sep. 2024 – .)

Alumni:

Zheyang Luo (M.Sc. 2026): Quantification and Prediction of Stock Market Investment Risk Based on Large-Scale Models; Jiahao Xiao (M.Sc. 2026): Federated Distillation for Multi-Domain Text Data and Out-of-Distribution; Xuankang Zhang (M.Sc. 2026): Research on Named Entity Recognition Methods for Large Language Models in Low-Resource Scenarios

RESEARCH SEMINAR

Invited Talks:

Oct. 2023	Beijing University of Posts and Telecommunications	Text Understanding and Generation in DRT
Oct. 2023	Peking University	Text Understanding and Generation in DRT
Jun. 2023	Yunnan University	Text Understanding and Generation in DRT
Apr. 2023	Keynote at CCF-NLP	Text Understanding and Generation in DRT

PROFESSIONAL SERVICE

Reviewer for journals:

Springer Nature: Language Resources and Evaluation, Computational Linguistics. ACM: Transactions on Asian and Low-Resource Language Information Processing. IEEE: Transactions on Big Data; Transactions on Audio, Speech and Language Processing; Transactions on Mobile Computing.

Reviewer for conferences:

ACL (2018, 19, 20, 21, 22, 23, 24, 25[AC], 26[AC]); EMNLP (2018, 19, 20, 21, 22, 23, 24, 25[AC], 26[AC]); NAACL (2021, 22, 23, 24, 26[AC]); COLING (18, 20, 22[AC], 24, 25, 26[AC]); EACL (2019, 21); CoNLL (2024, 25, 26); AAAI (2023, 24, 25, 26, 27); ECCV (2026); ICML (2026); NeuriPS (2026); CVPR (2026); CIKM (2026).

Chair of Workshops:

ACL student research workshop (2020).

PUBLICATIONS

Journal articles:

Jiahui Wang, **Jiangming Liu**, Kun Yue, Liang Duan, Jiande Wu. Structural Entropy Guided Relation Extraction on Adaptive Graph Structure. *Pattern Recognition*. 2025.

Jiangming Liu, Shay B Cohen, Mirella Lapata, Johan Bos. Universal Discourse Representation Structure Parsing. *Computational Linguistics*. 2021.

Jiangming Liu, Yue Zhang. In-order transition-based constituent parsing. *Transactions of the Association for Computational Linguistics*. 2017.

Jiangming Liu, Yue Zhang. Shift-Reduce Constituent Parsing with Neural Lookahead Features.

Transactions of the Association for Computational Linguistics. 2017.

JinAn Xu, **Jiangming Liu**, Yujie Zhang. Word Similarity Computing Based on Hybrid Hierarchical Structure by HowNet. *Information Science and Engineering*. 2015.

Jiangming Liu, JinAn Xu, Yujie Zhang. Summarization based on hidden topic Markov model with multi-features. *Acta Scientiarum Naturalium Universitatis Pekinensis*. 2014.

Pre-print:

Jiahao Xiao, **Jiangming Liu**. Out-of-Distribution Federated Distillation with Domain-Aware Proxy. *arXiv preprint arXiv:2608.08525*. 2026.

Xuankang Zhang, **Jiangming Liu**. DE-NER : Zero-shot Named Entity Recognition via Dialogue Elicitation of Large Language Models. *arXiv preprint arXiv:2608.00538*. 2026.

Yanjie Gou, **Jiangming Liu**, Kouying Xue, Yi Hu. Large-scale User Game Lifecycle Representation Learning. *arXiv preprint arXiv:2510.15412*. 2025.

Conference proceedings:

Guangbin Zhang, Zheyang Luo, **Jiangming Liu**. Adaptive-Margin Masking and Restoration for Balanced Multimodal Learning. In *The Fortieth Annual Conference on Neural Information Processing Systems (NeurIPS)*. 2026.

Guanheng Qu, Jingxiang Qiu, Na Li, **Jiangming Liu**. SNEU-Guide: Traceable Evidence-State Evaluation for Chinese Tour-Guide Training. In *Procceedings of The 15th CCF International Conference on Natural Language Processing and Chinese Computing (NLPCC)*. 2026.

Jingxiang Qiu, Guanheng Qu, Jianrui Li, **Jiangming Liu**. HybridRAFT: A Traffic-Oriented Hybrid Retrieval-Augmented Forecasting Network for Large-Scale Networked Traffic Flow Prediction. In *Proceedings of the 2026 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*. 2026.

Guangheng Qu, Fan Jiang, **Jiangming Liu**. SLRNet: Super Lightweight Residual Network for Real-Time Image Dehazing. In *Proceedings of 2026 International Conference on Intelligent Computing (ICIC)*. 2026.

Xuankang Zhang, **Jiangming Liu**. Named Entity Recognition in COVID-19 tweets with Entity Knowledge Augmentation. In *Proceedings of the 8th International Conference on Natural Language Processing (ICNLP)*. 2026.

Yulin Dou, **Jiangming Liu**. TO-GATE: Clarifying Questions and Summarizing Responses with Trajectory Optimization for Eliciting Human Preference. In *Proceedings of the AAAI Conference on Artificial Intelligence*. 2026.

Jiahao Xiao, **Jiangming Liu**. Adaptive Federated Distillation for Multi-Domain Non-IID Textual Data. In *Proceedings of International Joint Conference on Neural Networks (IJCNN)*. 2025.

Zheyang Luo, **Jiangming Liu**. QuanSIRA: The Quantitative Investment Risk Modeling in Stock Markets with Pre-trained Language Models. In *Proceedings of International Joint Conference on Neural Networks (IJCNN)*. 2025.

Zheyang Luo, Guangbin Zhang, Jiahao Xiao, Xuankang Zhang, Yulin Dou, **Jiangming Liu**. FMD-Mllama at the Financial Misinformation Detection Challenge Task: Multimodal Reasoning and Evidence Generation. In *Proceedings of the Joint Workshop of the 9th Financial Technology and Natural Language Processing, the 6th Financial Narrative Processing, and the 1st Workshop on Large Language Models for Finance and Legal*. 2025

Yanjie Gou, Yuanzhou Yao, Zhao Zhang, Yiqing Wu, Yi Hu, Fuzhen Zhuang, **Jiangming Liu**, Yongjun Xu. Controllable Multi-Behavior Recommendation for In-Game Skins with Large Sequential Model. In *Proceedings of the 30th Conference on Knowledge Discovery and Data Mining (SIGKDD)*. 2024

Jiangming Liu. Model-agnostic cross-lingual training for discourse representation structure parsing. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (COLING-LREC)*. 2024.

Jiangming Liu. Soft well-formed semantic parsing with score-based selection. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and*

Evaluation (COLING-LREC). 2024.

Hang Du, Sicheng Zhang, Binzhu Xie, Guoshun Nan, Jiayang Zhang, Junrui Xu, Hangyu Liu, Sicong Leng, **Jiangming Liu**, Hehe Fan, Dajiu Huang, Jing Feng, Linli Chen, Can Zhang, Xuhuan Li, Hao Zhang, Jianhang Chen, Qimei Cui, Xiaofeng Tao. Uncovering What, Why and How: A Comprehensive Benchmark for Causation Understanding of Video Anomaly. In *Proceedings of the Conference on Computer Vision and Pattern Recognition (CVPR)*. 2024.

Xinge Ma, **Jiangming Liu**, Jin Wang, Xuejie Zhang. Fedid: Federated interactive distillation for large-scale pretraining language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. **Outstanding Paper Award**. 2023

Jianyong Chen, **Jiangming Liu**, Jin Wang, Xuejie Zhang. Semantic candidate retrieval for few-shot entity linking. In *Proceedings of International Conference on Natural Language Processing and Chinese Computing (NLPCC)*. 2023

Wenjie Li, Qiaolin Xia, Junfeng Deng, Hao Cheng, **Jiangming Liu**, Kouying Xue, Yong Cheng, Shu-Tao Xia. VFed-SSD: Towards Practical Vertical Federated Advertising. In *Proceedings of the Trustworthy Federated Learning workshop of International Joint Conference on Artificial Intelligence (FL-IJCAI)*. 2022.

Jiangming Liu, Shay B Cohen, Mirella Lapata. Text generation from discourse representation structures. In *Proceedings of 2021 Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*. 2021.

Jiangming Liu, Shay B Cohen, Mirella Lapata. Dscorer: A fast evaluation metric for discourse representation structure parsing. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*. 2020.

Qiankun Fu, Yue Zhang, **Jiangming Liu**, Meishan Zhang. DRTS Parsing with Structure-Aware Encoding and Decoding. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*. 2020.

Matt Gardner, Yoav Artzi, Victoria Basmov, Jonathan Berant, Ben Bogin, Sihao Chen, Pradeep Dasigi, Dheeru Dua, Yanai Elazar, Ananth Gottumukkala, Nitish Gupta, Hannaneh Hajishirzi, Gabriel Ilharco, Daniel Khashabi, Kevin Lin, **Jiangming Liu**, Nelson F. Liu, Phoebe Mulcaire, Qiang Ning, Sameer Singh, Noah A. Smith, Sanjay Subramanian, Reut Tsarfaty, Eric Wallace, Ally Zhang, Ben Zhou. Evaluating Models' Local Decision Boundaries via Contrast Sets. In *Findings of the Association for Computational Linguistics (EMNLP)*. 2020.

Jiangming Liu, Matt Gardner, Shay B. Cohen, Mirella Lapata. Multi-Step Inference for Reasoning Over Paragraphs. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. 2020.

Jiangming Liu, Shay B Cohen, Mirella Lapata. Discourse representation parsing for sentences and documents. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL)*. 2019

Jiangming Liu, Shay B Cohen, Mirella Lapata. Discourse representation structure parsing with recurrent neural networks and the transformer model. In *Proceedings of the shared task on semantic parsing (IWCS)*. **Winner System Award**. 2019.

Jiangming Liu, Shay B Cohen, Mirella Lapata. Discourse representation structure parsing. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (ACL)*. 2018

Qi Liu, Yue Zhang, **Jiangming Liu**. Learning domain representation for multi-domain sentiment classification. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL)*. 2018.

Jiangming Liu, Yue Zhang. Encoder-Decoder Shift-Reduce Syntactic Parsing. In *Proceedings of the 15th International Conference on Parsing Technologies (IWPT)*. 2017.

Jiangming Liu, Yue Zhang. Attention modeling for targeted sentiment. In *Proceedings of the 15th Conference of the European Chapter of the Association for Computational Linguistics (EACL)*. 2017.

Jiangming Liu, Yue Zhang. An empirical comparison between n-gram and syntactic language

models for word ordering. In *Proceedings of the 2015 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. 2015.

JinAn Xu, **Jiangming Liu**, Yufeng Chen, Yujie Zhang, Fang Ming, Shaotong Li. Integrating Case Frame into Japanese to Chinese Hierarchical Phrase-based Translation Model. In *Proceedings of the 1st Workshop on Semantics-Driven Statistical Machine Translation (S2MT)*. 2015.

Jiangming Liu, JinAn Xu, Jun Xie, Yujie Zhang. Case Frame Constraints for Hierarchical Phrase-Based Translation: Japanese-Chinese as an Example. In *Proceedings of Natural Language Processing and Chinese Computing (NLPCC)*. 2014.

Jiangming Liu, Jinan Xu, Yujie Zhang. An approach of hybrid hierarchical structure for word similarity computing by HowNet. In *Proceedings of the Sixth International Joint Conference on Natural Language Processing (IJCNLP)*. 2013.